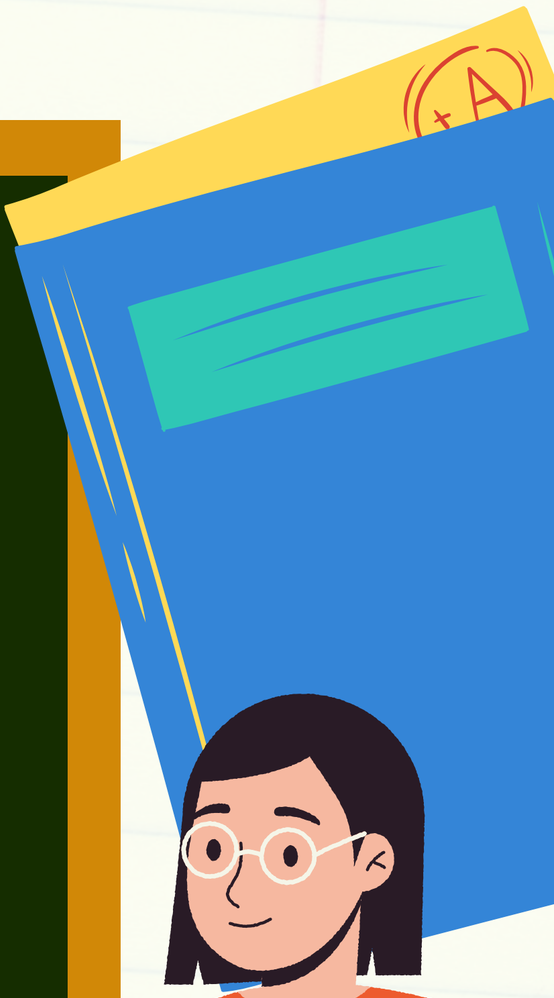
# Group Project

Presentation By Query Ninjas



# Members Of Group

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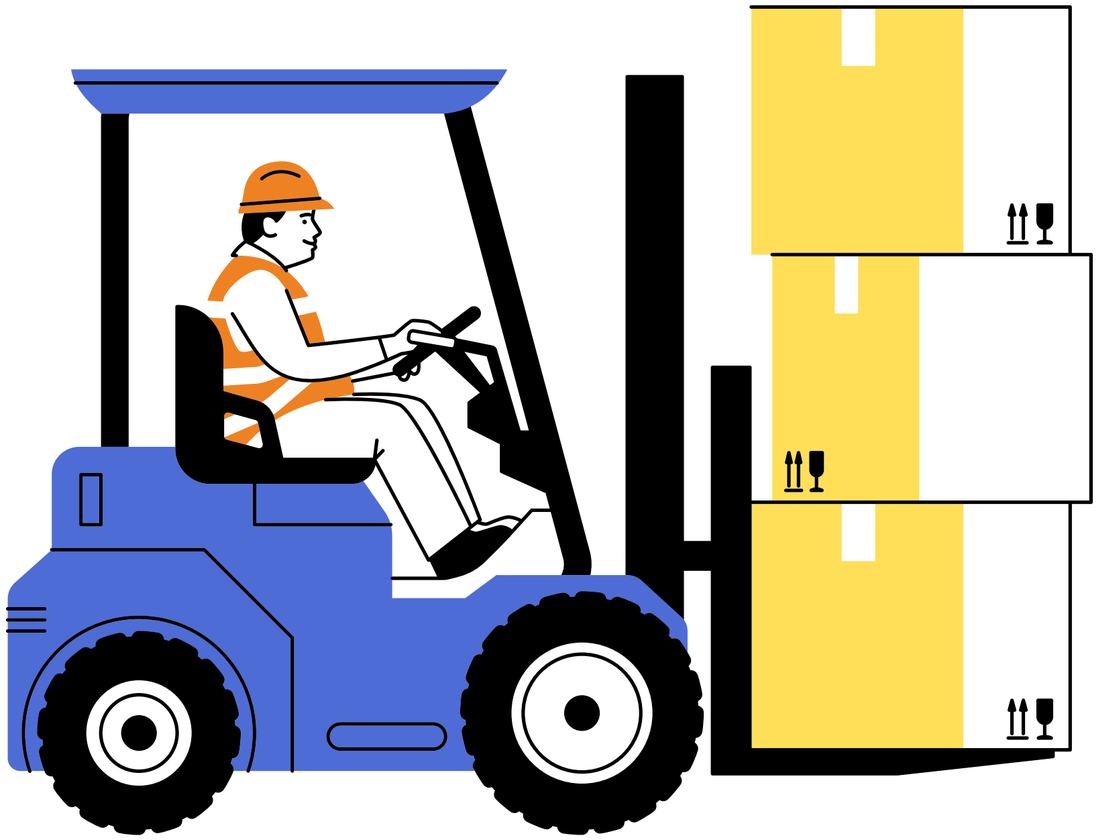
# Project Topic

## “Supply Chain Optimization and Insights using SQL Queries”

with Strategic Supply Chain Management

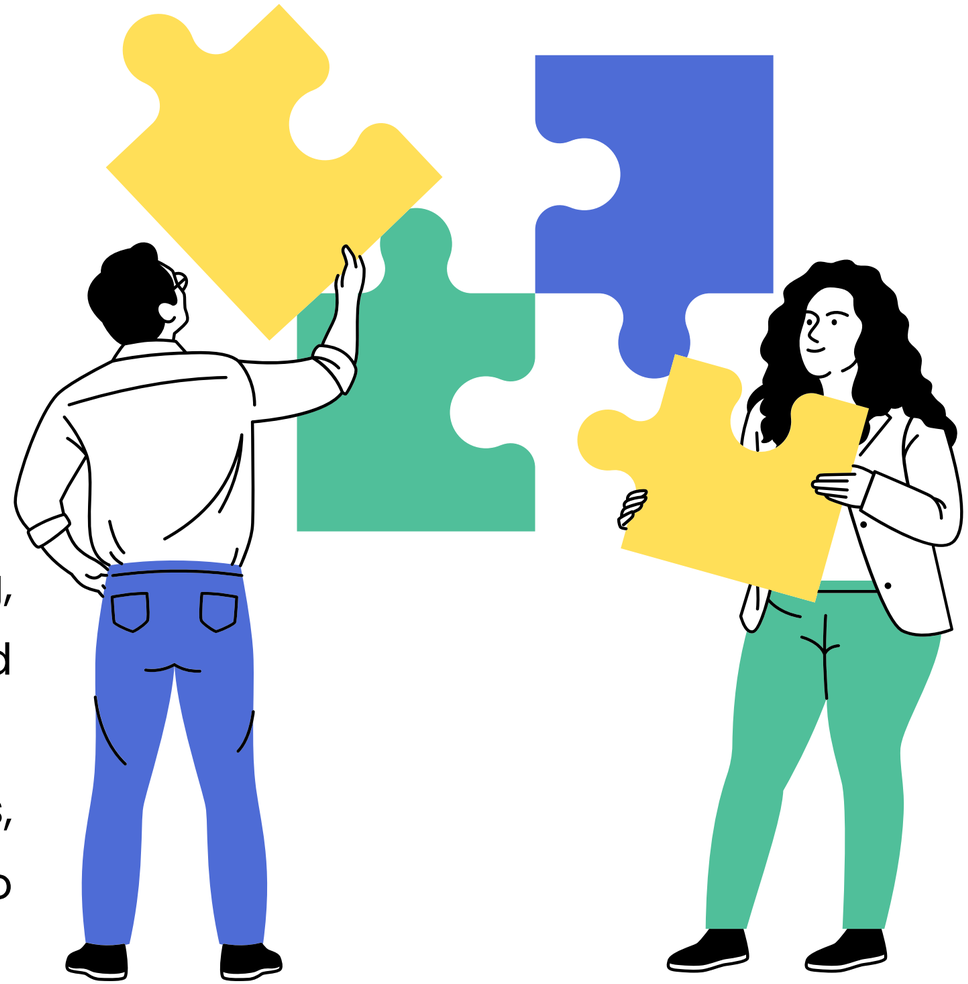
# SUPPLY CHAIN



MANAGEMENT

# INTRODUCTION TO SUPPLY CHAIN

## Management



Supply Chain Management (SCM) is the process of planning, managing, and controlling the flow of goods, services, and information from the supplier to the final customer.

It includes all activities involved in sourcing raw materials, manufacturing products, storing them, and delivering them to customers efficiently.

# KEY COMPONENTS OF SUPPLY CHAIN

## Management

1. Planning

Planning means deciding how much product to make and when to make it based on demand.

2. Sourcing (Procurement)

In this step, the company buys raw materials from suppliers.

3. Manufacturing (Production)

Raw materials are converted into finished products.

Example: cloth is turned into shirts.

Borcelle Company

# KEY COMPONENTS OF SUPPLY CHAIN

## Management

4. Inventory Management

Managing stock properly.

Not too much (waste of money) and not too little (shortage).

5. Warehousing

Storing finished products safely in a warehouse before delivery.

6. Transportation (Logistics)

Moving goods from one place to another.

Example: factory → warehouse → customer.

Borcelle Company

# KEY COMPONENTS OF SUPPLY CHAIN

## Management

7. Distribution

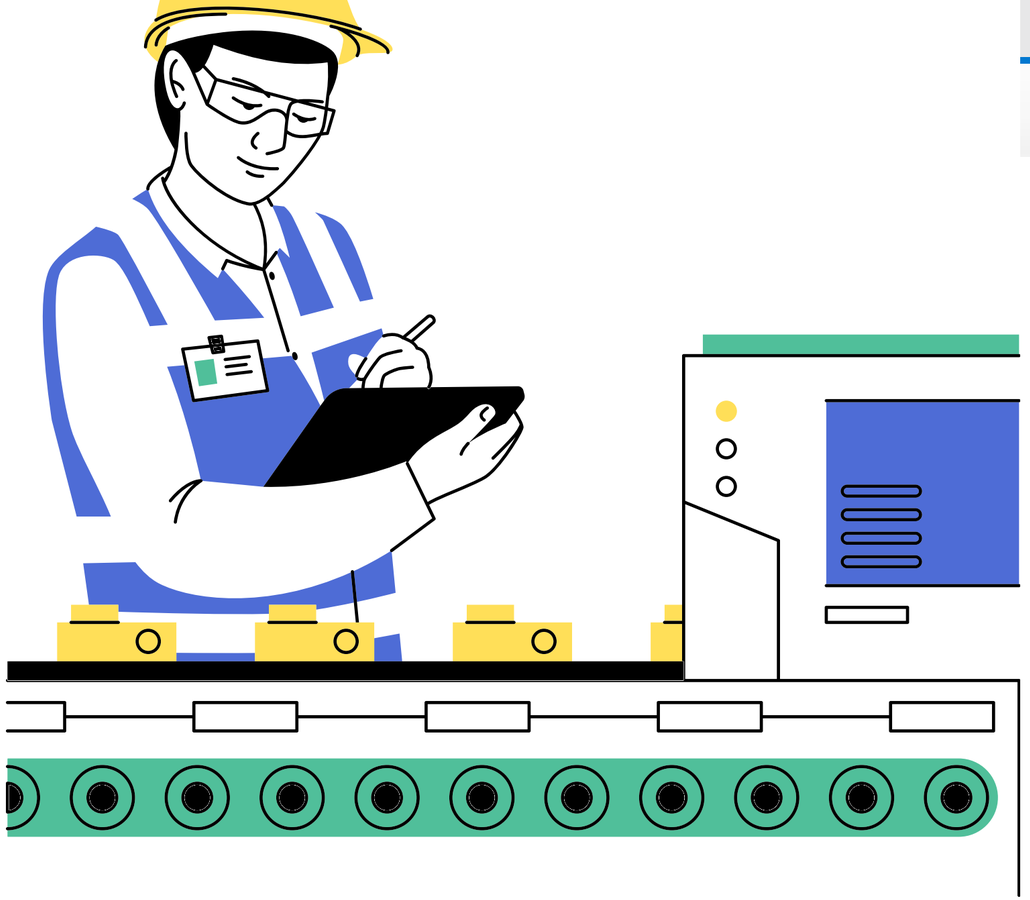
Delivering products to customers or retailers.

This is the final step.

8. Returns (Reverse Logistics)

Handling returned or defective products from customers.

“Supply Chain Management includes planning, sourcing, production, storage, transportation, and delivering products to customers efficiently.”



Full_Database_Setup* | sql project*

```
1    -- 1 Data Exploration: Read al
2 •  select * from customer;
3 •  select * from supplier;
4 •  select * from product;
5 •  select * from orderitem;
6 •  select * from orders;
```

"This query is used to display all the data from the main tables like customers, suppliers, products, order items, and orders. It helps us understand the structure and content of the database."

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| Id | CompanyName | ContactName | ContactTitle | City | Country | Phone | Fax |
|---|---|---|---|---|---|---|---|
| 1 | Exotic Liquids | Charlotte Cooper | NULL | London | UK | (171) 555-2222 | NULL |
| 2 | New Orleans Cajun Delights | Shelley Burke | NULL | New Orleans | USA | (100) 555-4822 | NULL |
| 3 | Grandma Kelly's Homestead | Regina Murphy | NULL | Ann Arbor | USA | (313) 555-5735 | (313) 555-3349 |
| 4 | Tokyo Traders | Yoshi Nagase | NULL | Tokyo | Japan | (03) 3555-5011 | NULL |
| 5 | Cooperativa de Quesos 'Las Cabras' | Antonio del Valle Saavedra | NULL | Oviedo | Spain | (98) 598 76 54 | NULL |
| 6 | Mayumi's | Mayumi Ohno | NULL | Osaka | Japan | (06) 431-7877 | NULL |
| 7 | Pavlova, Ltd. | Ian Devling | NULL | Melbourne | Australia | (03) 444-2343 | (03) 444-6588 |

customer 137 | supplier 138 | product 139 | orderitem 140 | orders 141

```sql
select country,count(id) count_customer
from customer group by country
order by count_customer desc;
```

Result Grid | Filter Rows:

| country | count_customer |
|---|---|
| USA | 13 |
| Germany | 11 |
| France | 11 |
| Brazil | 9 |
| UK | 7 |
| Mexico | 5 |
| Spain | 5 |

Result 143

“This query counts the number of customers in each country and sorts them from highest to lowest. It helps to understand which country has the most customers.”

```sql
select * from product where IsDiscontinued="1";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

| Id | ProductName | SupplierId | UnitPrice | Package | IsDiscontinued |
|---|---|---|---|---|---|
| 5 | Chef Anton's Gumbo Mix | 2 | 21.35 | 36 boxes | 1 |
| 9 | Mishi Kobe Niku | 4 | 97.00 | 18 - 500 g pkgs. | 1 |
| 17 | Alice Mutton | 7 | 39.00 | 20 - 1 kg tins | 1 |
| 24 | Guaraná Fantástica | 10 | 4.50 | 12 - 355 ml cans | 1 |
| 28 | Rössle Sauerkraut | 12 | 45.60 | 25 - 825 g cans | 1 |
| 29 | Thüringer Rostbratwurst | 12 | 123.79 | 50 bags x 30 sausgs. | 1 |
| 42 | Singaporean Hokkien Fried Mee | 20 | 14.00 | 32 - 1 kg pkgs. | 1 |

product 144

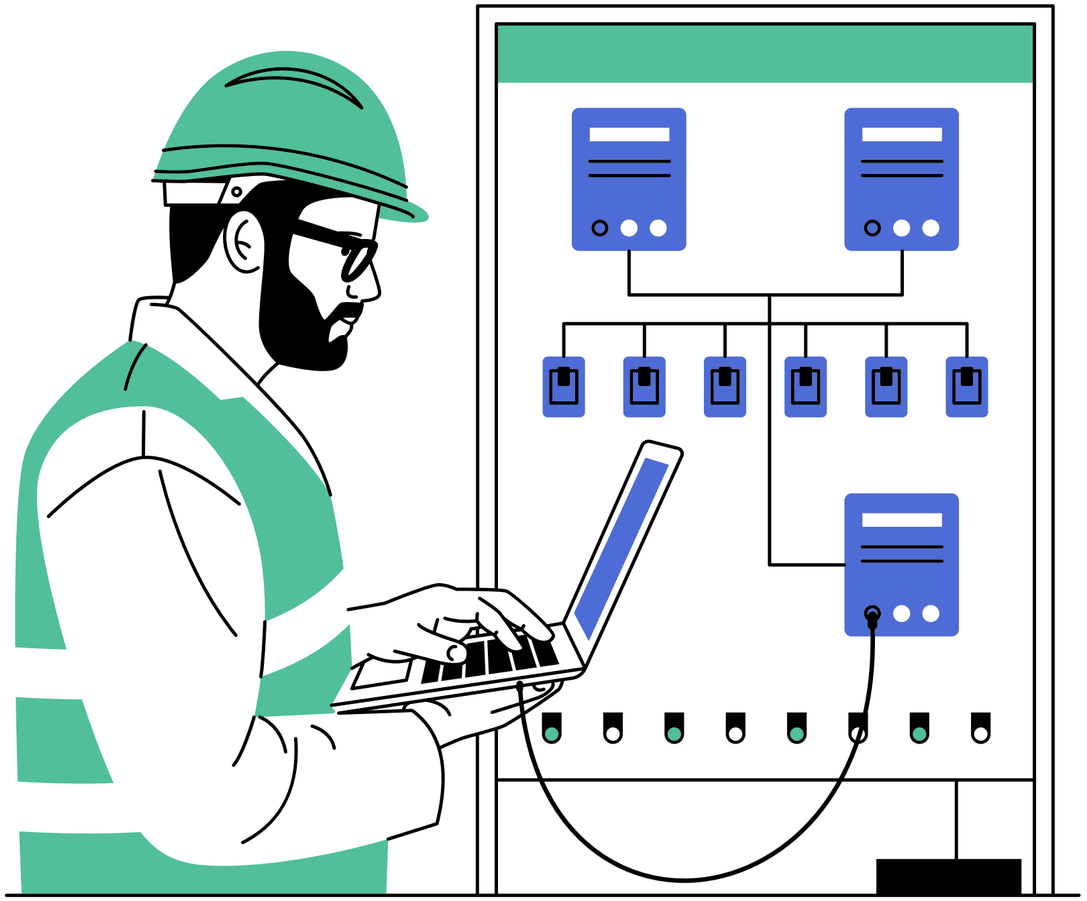
“This query shows all products that are currently active and not discontinued. It helps to identify available products for sale.”

```sql
select s.companyname, p.productname
from supplier s
join product p on s.id = p.supplierid;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

| companyname | productname |
|---|---|
| Aux joyeux ecclésiastiques | Côte de Blaye |
| Aux joyeux ecclésiastiques | Chartreuse verte |
| Bigfoot Breweries | Sasquatch Ale |
| Bigfoot Breweries | Steeleye Stout |
| Bigfoot Breweries | Laughing Lumberjack Lager |
| Cooperativa de Quesos 'Las Cabras' | Queso Cabrales |
| Cooperativa de Quesos 'Las Cabras' | Queso Manchego La Pastora |

Result 145



“This query is intended to display supplier company names along with the products they supply.”

```
select  * from customer where
country = 'Mexico';
```

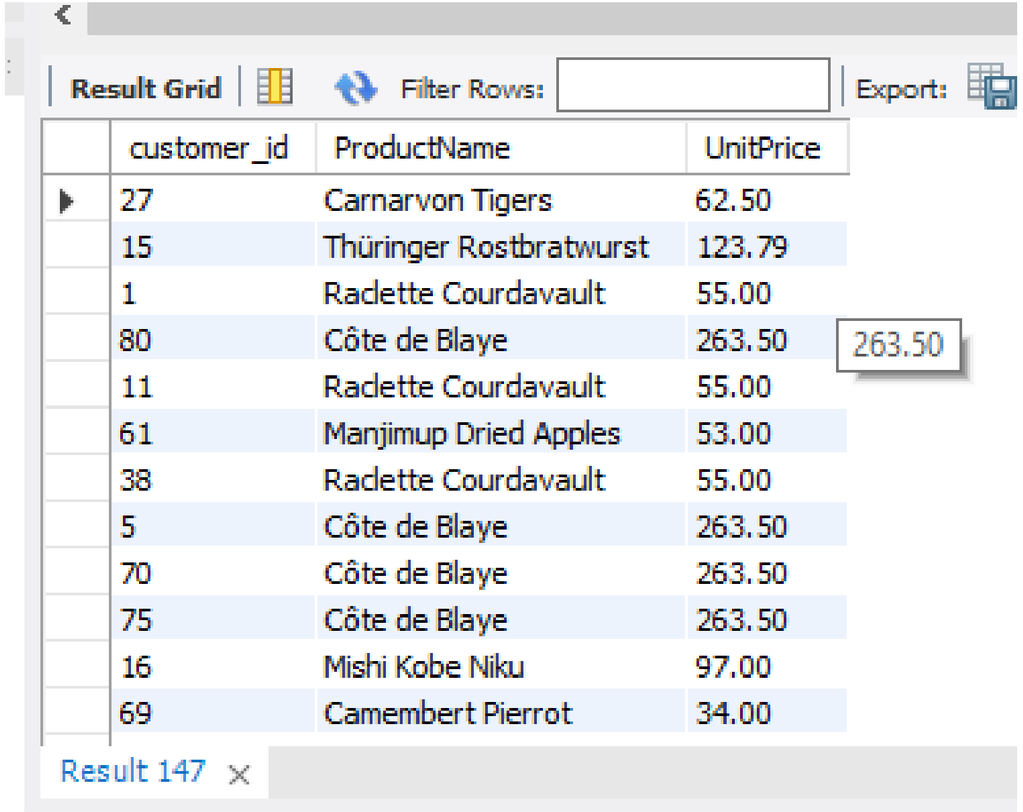
Result Grid | Filter Rows: | Edit: | Expor

| Id | FirstName | LastName | City | Country | Phone |
|---|---|---|---|---|---|
| 2 | Ana | Trujillo | México D.F. | Mexico | (5) 555-4729 |
| 3 | Antonio | Moreno | México D.F. | Mexico | (5) 555-3932 |
| 13 | Francisco | Chang | México D.F. | Mexico | (5) 555-3392 |
| 58 | Guillermo | Fernández | México D.F. | Mexico | (5) 552-3745 |
| 80 | Miguel | Angel Paolino | México D.F. | Mexico | (5) 555-2933 |
| NULL | NULL | NULL | NULL | NULL | NULL |

customer 146

“This query retrieves all customer details who are located in Mexico. It helps in region-based analysis.”

```sql
select c.id as customer_id,p.ProductName,p
from customer c
join orders o
on c.id=o.CustomerId
join orderitem oi
on o.id=oi.OrderId
join product p
on p.id=oi.ProductId
where (c.id,p.UnitPrice) in
(
select c.id,max(p.UnitPrice)
from customer c
join orders o
on o.CustomerId =c.Id
join orderitem oi
on oi.OrderId=o.Id
join product p
on p.Id=oi.ProductId
group by c.Id);
```



Result Grid | Filter Rows: | Export:

| customer_id | ProductName | UnitPrice |
|---|---|---|
| 27 | Carnarvon Tigers | 62.50 |
| 15 | Thüringer Rostbratwurst | 123.79 |
| 1 | Raclette Courdavault | 55.00 |
| 80 | Côte de Blaye | 263.50 |
| 11 | Raclette Courdavault | 55.00 |
| 61 | Manjimup Dried Apples | 53.00 |
| 38 | Raclette Courdavault | 55.00 |
| 5 | Côte de Blaye | 263.50 |
| 70 | Côte de Blaye | 263.50 |
| 75 | Côte de Blaye | 263.50 |
| 16 | Mishi Kobe Niku | 97.00 |
| 69 | Camembert Pierrot | 34.00 |

263.50

Result 147

“This query finds the most expensive product ordered by each customer. It helps to identify high-value purchases made by customers.”

```sql
with max_p as(select s.id,count(p.id) as cnt_prod
from supplier s
join product p
on p.SupplierId=s.Id
group by s.id)
select * from max_p
where cnt_prod in (select max(cnt_prod) from max_p);
```

Result Grid | Filter Rows:

| id | cnt_prod |
|---|---|
| 7 | 5 |
| 12 | 5 |

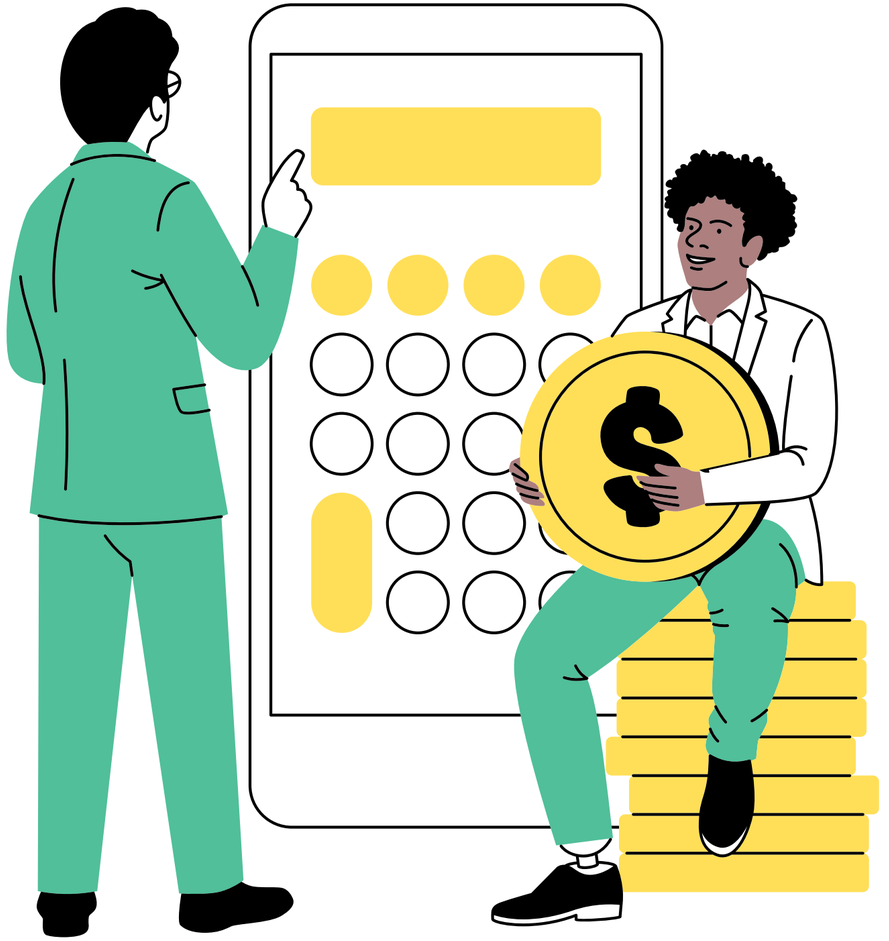
“This query identifies the supplier who provides the highest number of products. It helps to find the most active supplier.”

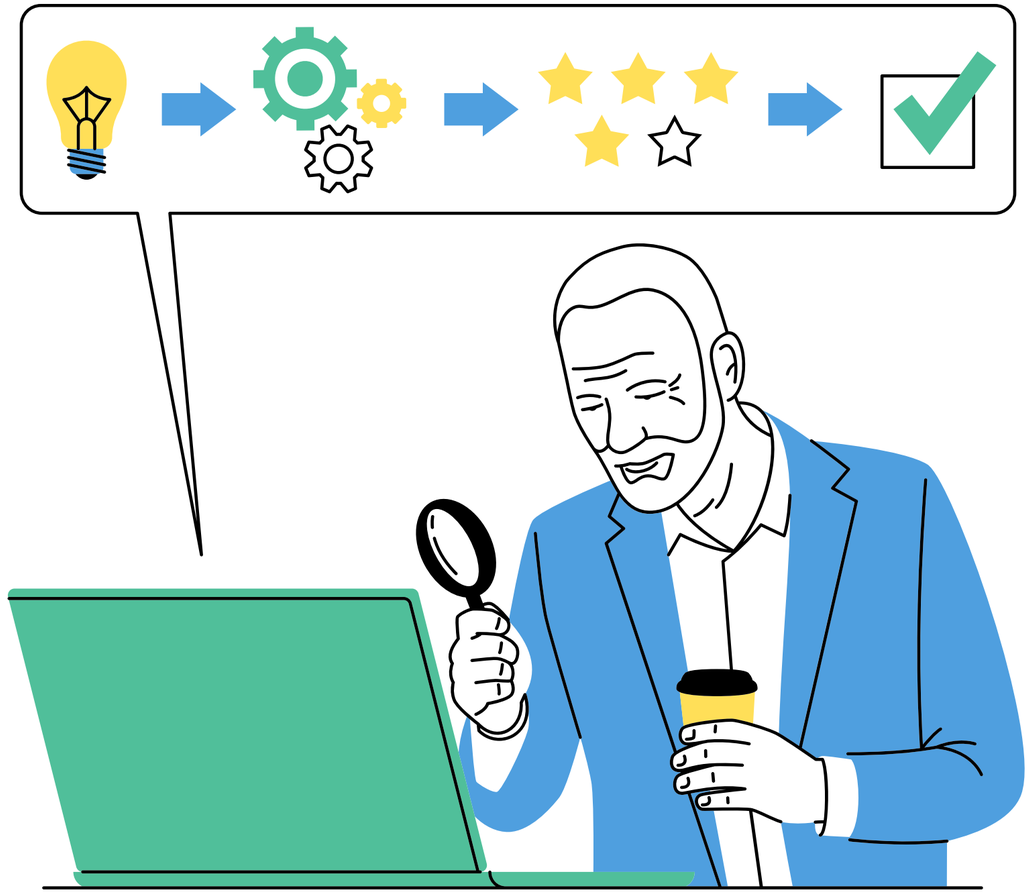
```sql
select year(orderdate) as year,month(orderdate) as month, count(id) as coount_order
from orders
group by year(orderdate),month(orderdate)
order by year asc,month asc;
```

Result Grid | Filter Rows:

| year | month | coount_order |
|---|---|---|
| NULL | NULL | 830 |

“This query counts the number of orders placed each month and year. It helps to analyze order trends over time.”





```
with max_b as(select country, count(id)
over (partition by country) as count
from supplier)
select distinct * from max_b
where count in (select max(count) from max_b);
```

Result Grid | Filter Rows:

| country | count |
|---|---|
| USA | 4 |

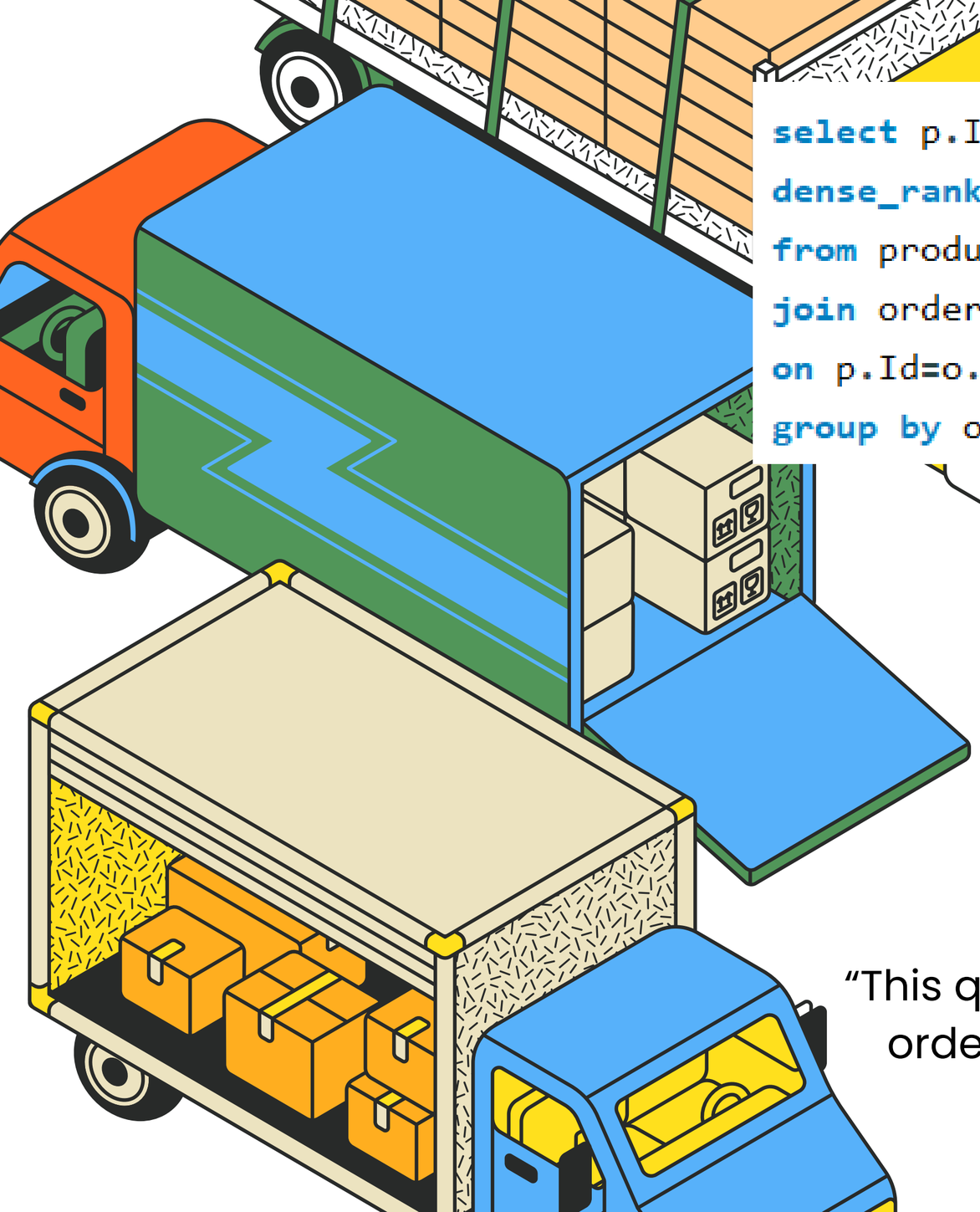
“This query finds the country that has the maximum number of suppliers. It helps in understanding supplier distribution.”

```
select * from customer
where Id not in
(select CustomerId from orders);
```

Result Grid | Filter Rows: | Edit:

| Id | FirstName | LastName | City | Country | Phone |
|---|---|---|---|---|---|
| 22 | Diego | Roel | Madrid | Spain | (91) 555 94 44 |
| 57 | Marie | Bertrand | Paris | France | (1) 42.34.22.66 |
| NULL | NULL | NULL | NULL | NULL | NULL |

“This query identifies customers who have never placed any orders. It helps to find inactive customers.”



```sql
select p.Id,p.ProductName,sum(o.quantity) as total_quantity,
dense_rank() over(order by sum(o.quantity) desc) as product_rank
from product p
join orderitem o
on p.Id=o.ProductId
group by o.Id,p.ProductName;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

| Id | ProductName | total_quantity | product_rank |
|---|---|---|---|
| 39 | Chartreuse verte | 130 | 1 |
| 64 | Wimmers gute Semmelknödel | 130 | 1 |
| 55 | Pâté chinois | 120 | 2 |
| 55 | Pâté chinois | 120 | 2 |
| 27 | Schoggi Schokolade | 120 | 2 |
| 61 | Sirop d'érable | 120 | 2 |
| 41 | Jack's New England Clam Chowder | 120 | 2 |
| 53 | Perth Pasties | 120 | 2 |

“This query ranks products based on the total quantity ordered by customers. It helps to identify the most demanded products.”

```sql
select c.*,sum(TotalAmount) total_spent
from customer c
join orders o
on c.id=o.CustomerId
group by c.id
order by sum(TotalAmount) desc limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

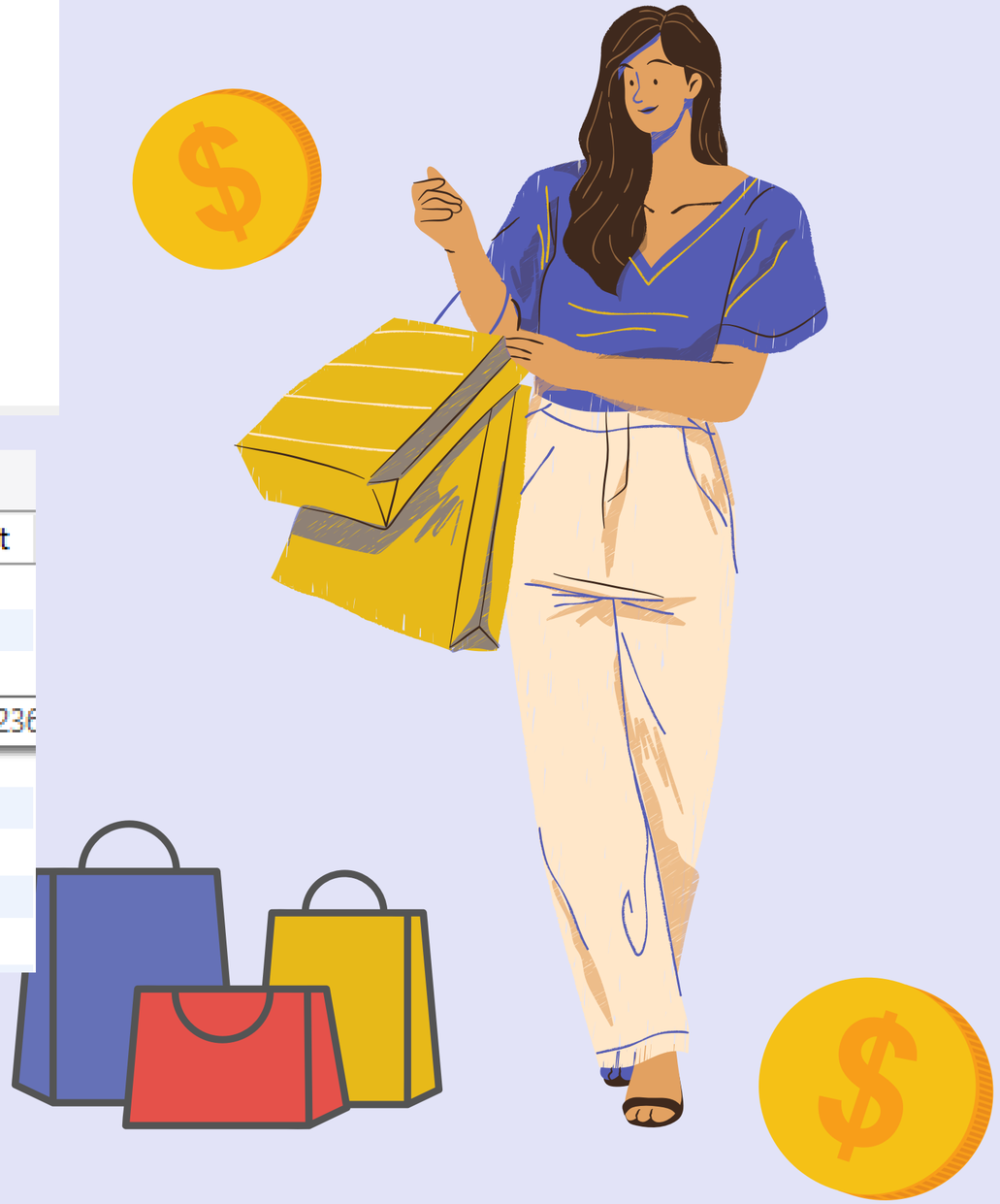
| Id | FirstName | LastName | City | Country | Phone | total_spent |
|---|---|---|---|---|---|---|
| 63 | Horst | Kloss | Cunewalde | Germany | 0372-035188 | 117483.39 |

"This query finds the customer who has spent the most money across all orders. It helps to identify high-value customers."

```sql
select c.*,sum(TotalAmount) as total_order_amount
from customer c
join orders o
on c.Id=o.CustomerId
group by c.id
order by sum(TotalAmount) desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Id | FirstName | LastName | City | Country | Phone | total_order_amount |
|---|---|---|---|---|---|---|
| 63 | Horst | Kloss | Cunewalde | Germany | 0372-035188 | 117483.39 |
| 71 | Jose | Pavarotti | Boise | USA | (208) 555-8097 | 115673.39 |
| 20 | Roland | Mendel | Graz | Austria | 7675-3425 | 113236.68 |
| 37 | Patricia | McKenna | Cork | Ireland | 2967 542 | 57317.39 |
| 65 | Paula | Wilson | Albuquerque | USA | (505) 555-5939 | 52245.90 |
| 34 | Mario | Pontes | Rio de Janeiro | Brazil | (21) 555-0091 | 34101.15 |
| 24 | Maria | Larsson | Bräcke | Sweden | 0695-34 67 21 | 32555.55 |
| 51 | Jean | Fresnière | Montréal | Canada | (514) 555-8054 | 32203.90 |
| 39 | Philip | Cramer | Brandenburg | Germany | 0555-09876 | 31745.75 |



“This query lists all customers based on their total spending, from highest to lowest. It helps in customer segmentation.”

```sql
select  c.id AS customer_id, c.firstname, o.OrderDate as current_dates,
lag(o.OrderDate) over (partition by c.id order by o.OrderDate) as previous_date
from customer c
join orders o ON o.CustomerId = c.id
order by c.id, o.OrderDate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

| customer_id | firstname | current_dates | previous_date |
|---|---|---|---|
| 1 | Maria | Apr 9 2014 12:00:00 | NULL |
| 1 | Maria | Aug 25 2013 12:00:00 | Apr 9 2014 12:00:00 |
| 1 | Maria | Jan 15 2014 12:00:00 | Aug 25 2013 12:00:00 |
| 1 | Maria | Mar 16 2014 12:00:00 | Jan 15 2014 12:00:00 |
| 1 | Maria | Oct 3 2013 12:00:00 | Mar 16 2014 12:00:00 |
| 1 | Maria | Oct 13 2013 12:00:00 | Oct |
| 2 | Ana | Aug 8 2013 12:00:00 | NULL |
| 2 | Ana | Mar 4 2014 12:00:00 | Aug 8 2013 12:00:00 |

Mar 16 2014 12:00:0

"This query shows current and previous order dates for each customer. It helps to analyze how frequently customers place orders."

```sql
select s.id as supplierid,s.ContactName,sum( oi.Quantity * oi.UnitPrice)
as total_revenue
from supplier s
join product p on s.id = p.supplierid
join orderitem oi on oi.productid = p.id
join orders o on o.id = oi.orderid
group by s.id,s.ContactName
order by  total_revenue desc limit 3;
```

Result Grid | Filter Rows:

| supplierid | ContactName | total_revenue |
|---|---|---|
| 18 | Guylène Nodier | 163135.00 |
| 12 | Martin Bein | 155946.55 |
| 28 | Eliane Noz | 126582.00 |

1

2

**“This query identifies the top 3 suppliers based on the total revenue generated from their products.”**

```sql
select customer_id, MAX(orderdate) AS latest_order_date
 from  (
   select o.customerid as customer_id,o.orderdate,
       row_number() over (partition by o.customerid
           order by o.orderdate) as rn
   from orders o) t
where rn > 1
group by customer_id;
```

| customer_id | latest_order_date |
|---|---|
| 1 | Oct 13 2013 12:00:00 |
| 2 | Sep 18 2012 12:00:00 |
| 3 | Sep 25 2013 12:00:00 |
| 4 | Oct 16 2013 12:00:00 |
| 5 | Sep 17 2013 12:00:00 |
| 6 | Mar 17 2014 12:00:00 |
| 7 | Sep 23 2013 12:00:00 |

Result 167

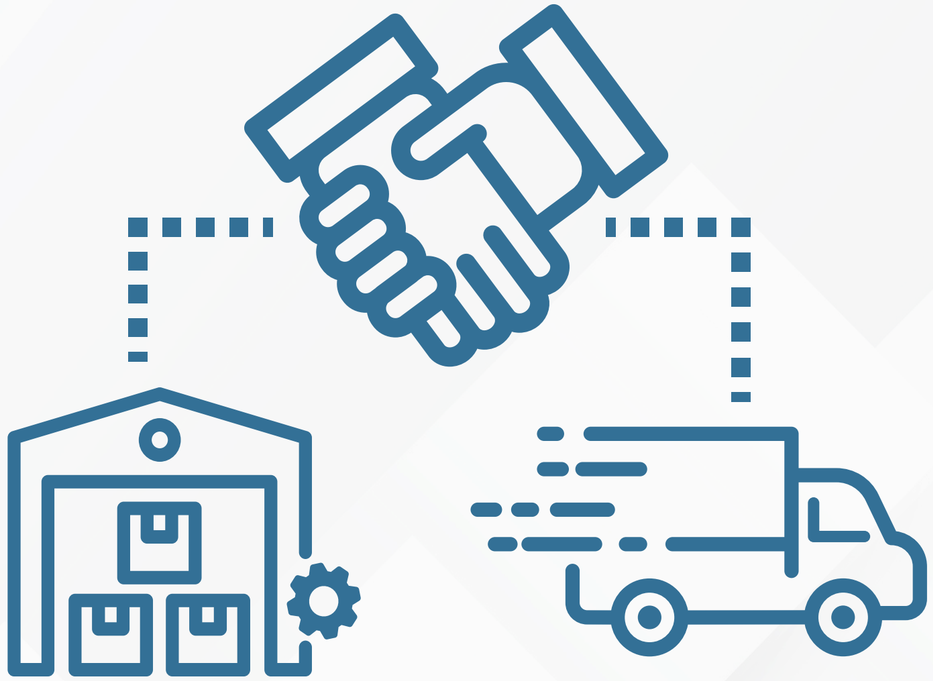
“This query shows the latest order date for each customer, excluding their first order. It helps to analyze repeat customers.”

```
select oi.OrderId,p.ProductName,s.ContactName
as Supplier_Name
from supplier s
join product p
on s.Id=p.SupplierId
join orderitem oi
on p.Id=oi.ProductId
join orders o
on o.Id=oi.OrderId;
```

Result Grid | Filter Rows:

| OrderId | ProductName | Supplier_Name |
|---|---|---|
| 38 | Chai | Charlotte Cooper |
| 47 | Chai | Charlotte Cooper |
| 70 | Chai | Charlotte Cooper |
| 101 | Chai | Charlotte Cooper |
| 107 | Chai | Charlotte Cooper |
| 123 | Chai | Charlotte Cooper |
| 159 | Chai | Charlotte Cooper |

Result 168



"This query displays order ID, product name, and supplier name for every order. It gives complete order information."

```
with product_order
as(select distinct c.*,count(c.id) as count
from customer c
join orders o
on c.Id=o.CustomerId
group by c.id)
select * from product_order
where count >10
order by count desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Id | FirstName | LastName | City | Country | Phone | count |
|---|---|---|---|---|---|---|
| 71 | Jose | Pavarotti | Boise | USA | (208) 555-8097 | 31 |
| 20 | Roland | Mendel | Graz | Austria | 7675-3425 | 30 |
| 63 | Horst | Kloss | Cunewalde | Germany | 0372-035188 | 28 |
| 24 | Maria | Larsson | Bräcke | Sweden | 0695-34 67 21 | 19 |
| 37 | Patricia | McKenna | Cork | Ireland | 2967 542 | 19 |
| 5 | Christina | Berglund | Luleå | Sweden | 0921-12 34 65 | 18 |
| 35 | Carlos | Hernández | San Cristóbal | Venezuela | (5) 555-1340 | 18 |

Result 169

“This query identifies customers who have ordered more than 10 products. It helps to find bulk buyers.”

```
select p.*
from product p
join orderitem oi
on p.Id=oi.ProductId
where Quantity=1;
```

"This query lists products that were ordered in a quantity of exactly one. It helps to identify small orders."

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Id | ProductName | SupplierId | UnitPrice | Package | IsDiscontinued |
|---|---|---|---|---|---|
| 37 | Gravad lax | 17 | 26.00 | 12 - 500 g pkgs. | 0 |
| 19 | Teatime Chocolate Biscuits | 8 | 9.20 | 10 boxes x 12 pieces | 0 |
| 69 | Gudbrandsdalsost | 15 | 36.00 | 10 kg pkg. | 0 |
| 13 | Konbu | 6 | 6.00 | 2 kg box | 0 |
| 72 | Mozzarella di Giovanni | 14 | 34.80 | 24 - 200 g pkgs. | 0 |
| 59 | Raclette Courdavault | 28 | 55.00 | 5 kg pkg. | 0 |
| 31 | Gorgonzola Telino | 14 | 12.50 | 12 - 100 g pkgs | 0 |

Result 170

```
select distinct c1.Id,c1.FirstName,c2.FirstName,c1.City,c1.country
from customer c1
join  customer c2
on (c1.City,c1.Country)= (c2.City,c2.Country)
where  c1.Id!=c2.Id
order by c1.country;
```

"This query finds customers who live in the same city and country. It helps to identify local customer groups."

Result Grid | Filter Rows: | Export: | Wrap

| Id | FirstName | FirstName | City | country |
|---|---|---|---|---|
| 64 | Sergio | Yvonne | Buenos Aires | Argentina |
| 12 | Patricio | Yvonne | Buenos Aires | Argentina |
| 12 | Patricio | Sergio | Buenos Aires | Argentina |
| 64 | Sergio | Patricio | Buenos Aires | Argentina |
| 54 | Yvonne | Patricio | Buenos Aires | Argentina |
| 54 | Yvonne | Sergio | Buenos Aires | Argentina |
| 21 | Aria | Pedro | Sao Paulo | Brazil |

Result 171

```sql
select oi.OrderId,sum((p.UnitPrice - oi.UnitPrice)* oi.Quantity) as total_saving,
rank () over(order by sum((p.UnitPrice - oi.UnitPrice)* oi.Quantity)desc) as order_rank
from product p
join orderitem oi
on p.Id=oi.ProductId
group by oi.OrderId;
```

Result Grid | Filter Rows:

| OrderId | total_saving | order_rank |
|---|---|---|
| 125 | 3072.40 | 1 |
| 177 | 2873.30 | 2 |
| 170 | 2820.80 | 3 |
| 106 | 2685.40 | 4 |
| 232 | 2625.30 | 5 |
| 113 | 1854.05 | 6 |
| 77 | 1542.55 | 7 |

Result 173

“This query calculates the total savings for each order and ranks them based on the highest savings. It helps to analyze discount impact.”

```sql
select p.productname, SUM(oi.quantity) as total_demand
from product p
join  orderitem oi on p.id = oi.productid
group by p.productname
order by total_demand desc;
```

“THIS QUERY IDENTIFIES HIGH-DEMAND PRODUCTS BASED ON TOTAL QUANTITY ORDERED. IT HELPS NEW SUPPLIERS CHOOSE POPULAR PRODUCTS.”

Result Grid | Filter Rows:

| productname | total_demand |
|---|---|
| Camembert Pierrot | 1577 |
| Raclette Courdavault | 1496 |
| Gorgonzola Telino | 1397 |
| Gnocchi di nonna Alice | 1263 |
| Pavlova | 1158 |
| Rhönbräu Klosterbier | 1155 |
| Guaraná Fantástica | 1125 |

Result 174

```
select s.contactname, p.productname
from orderitem oi
join product p on p.id = oi.productid
join supplier s on p.supplierid = s.id;
```

Result Grid | Filter Rows:

| contactname | productname |
|---|---|
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |
| Charlotte Cooper | Chai |

Result 175

**“THIS QUERY LISTS SUPPLIERS AND THE PRODUCTS THEY SUPPLY. IT HELPS TO IDENTIFY COMPETITORS IN THE MARKET.”**

# Thank You